Courier Management System (CMS)

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Courier Management System (CMS)

User Story

Courier and logistics organizations need to handle a large number of parcels, customers, delivery agents, branches, and billing activities every day. When these activities depend on manual records and separate systems, shipment tracking becomes difficult and delivery information may not reach customers on time. Manual assignment of delivery agents can also create workload conflicts and delays. In addition, the absence of automated notifications makes communication less effective, while limited reporting makes it difficult for managers to understand operational performance. Managing courier activities across different branches can further increase coordination problems. These issues can affect service quality, employee productivity, customer satisfaction, and organizational revenue.

The Courier Management System (CMS) provides a centralized Salesforce-based cloud solution for managing these activities. It brings shipment booking, customer management, delivery assignment, parcel tracking, billing, branch management, notifications, and performance analysis into one platform.

1. INTRODUCTION

1.1 Project Overview

Courier businesses handle continuous shipment and delivery operations that require accurate information and timely updates. Traditional manual methods can make it difficult to maintain shipment records, monitor parcel movement, assign delivery agents, and coordinate activities between branches.

The Courier Management System (CMS) is developed using the Salesforce platform to provide an integrated environment for courier operations. The system manages customer details, shipment records, delivery agents, branches, delivery updates, and invoices through Salesforce Objects.

Salesforce Flows are used to automate important processes, while Validation Rules help maintain accurate data. Reports and Dashboards provide managers with operational information and performance insights. The solution therefore provides centralized data management, improved tracking visibility, reduced manual work, and better coordination between different users.

1.2 Purpose

The purpose of the Courier Management System (CMS) is to provide a centralized Salesforce platform for managing courier activities in an organized and efficient manner.

The system is intended to simplify shipment creation, customer management, delivery agent assignment, parcel status updates, billing, and branch-level operations. It also uses Salesforce automation to reduce repetitive manual activities and maintain consistency in courier processes.

Through reports and dashboards, managers can monitor shipment progress, delivery activities, agent performance, and revenue-related information. The system ultimately aims to improve operational efficiency, provide better customer service, reduce processing errors, and support informed management decisions.

2. IDEATION PHASE

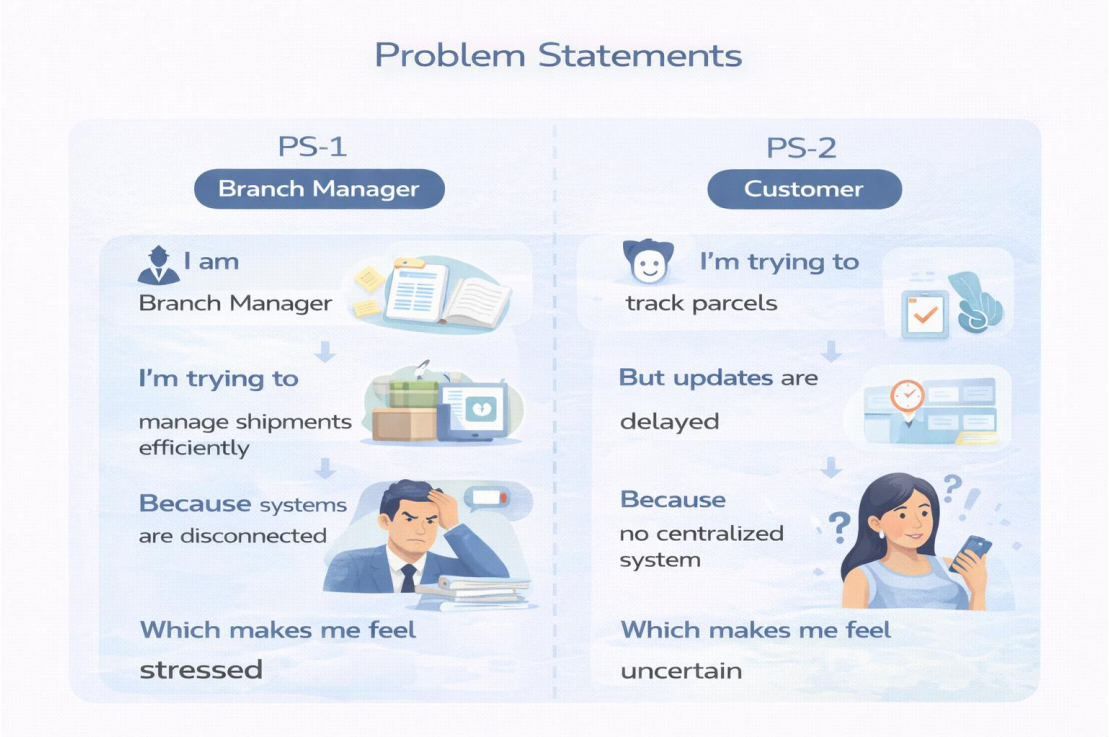
2.1 Problem Statement (Customer Problem Statement Template)

Courier operations involve several users with different responsibilities. Branch managers need to coordinate shipments and delivery activities, while customers need clear and timely information about their parcels.

PS-1 – Branch Manager:
I am a Branch Manager. I am trying to manage shipments and delivery operations efficiently, but shipment tracking and agent assignment are handled manually. Because courier information is maintained through disconnected processes, it becomes difficult to coordinate activities accurately, which makes me feel stressed about delays and operational errors.

PS-2 – Customer:
I am a Customer. I am trying to know the current location and status of my parcel, but shipment updates are not always available immediately. Because there is no centralized tracking mechanism, I may be uncertain about when my shipment will arrive.

PS-3 – Delivery Agent:
I am a Delivery Agent. I am trying to complete my assigned deliveries on time, but manual assignment and status updating can make daily operations difficult. Because delivery information is not centrally managed, I may face confusion regarding assigned shipments.



2.2 Empathy Map Canvas

User Persona: Branch Manager

Think & Feel:
Wants shipments to be delivered on time and expects accurate operational information.

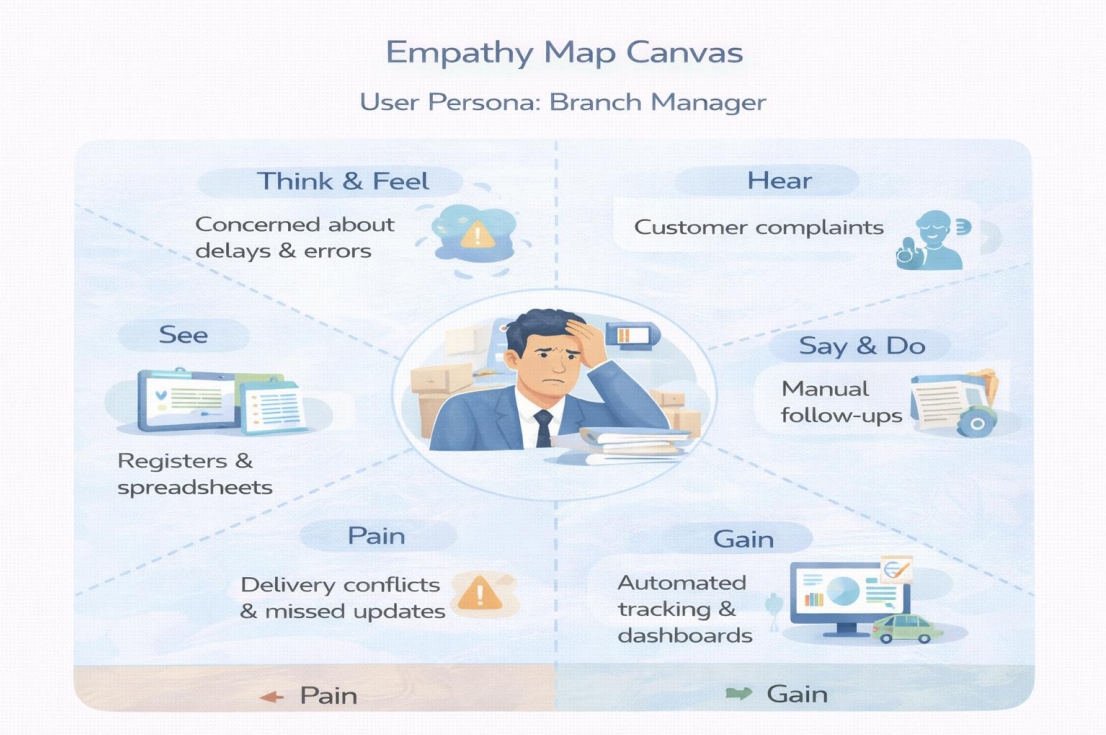
Hear:
Receives customer questions, delivery complaints, and updates from delivery agents.

See:
Manual records, spreadsheets, shipment information, and branch-level operational data.

Say & Do:
Follows up with agents, checks shipment status, coordinates branch activities, and resolves delivery issues.

Pain:
Delayed updates, manual tracking, agent coordination problems, and difficulty monitoring branch performance.

Gain:
Automated shipment tracking, centralized Salesforce records, quick status updates, and management dashboards.



2.3 Brainstorming & Idea Prioritization

Ideas Considered

Manual register-based management

Spreadsheet-based tracking

Independent courier web portal

Salesforce-based Courier Management System

Final Selection

The Salesforce-based CMS was selected because it provides:

Business process automation

Centralized cloud data

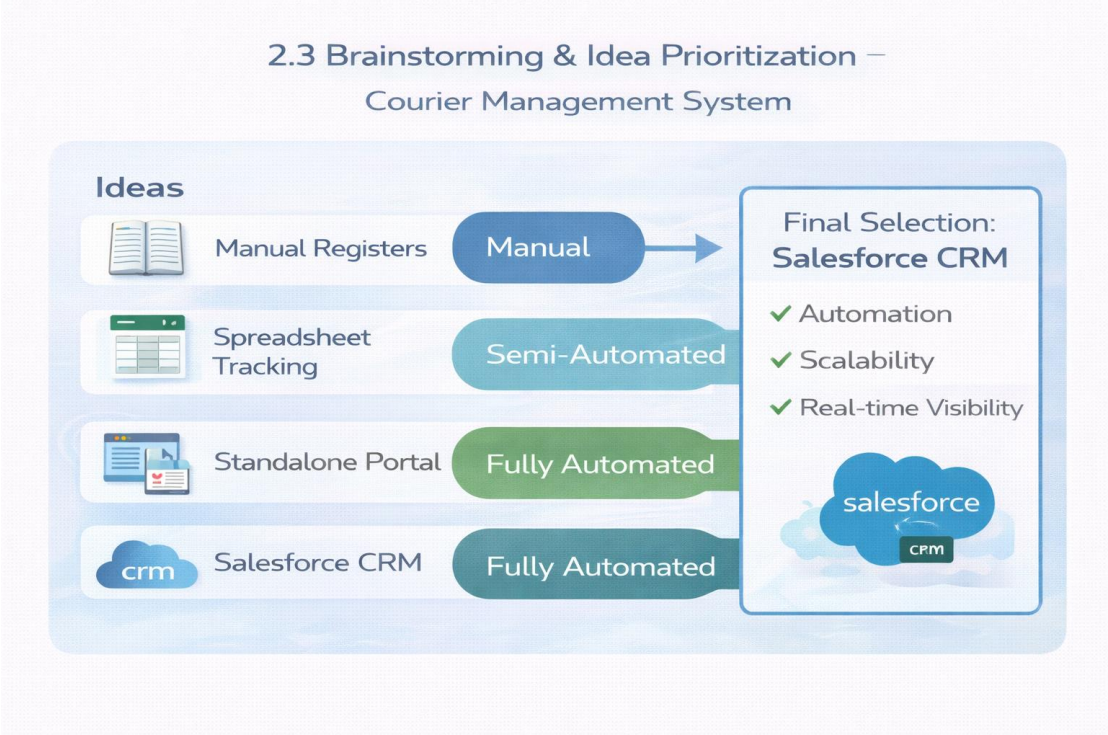
Real-time operational visibility

Custom objects and relationships

Validation and security controls

Reports and dashboards

Scalability for multiple branches



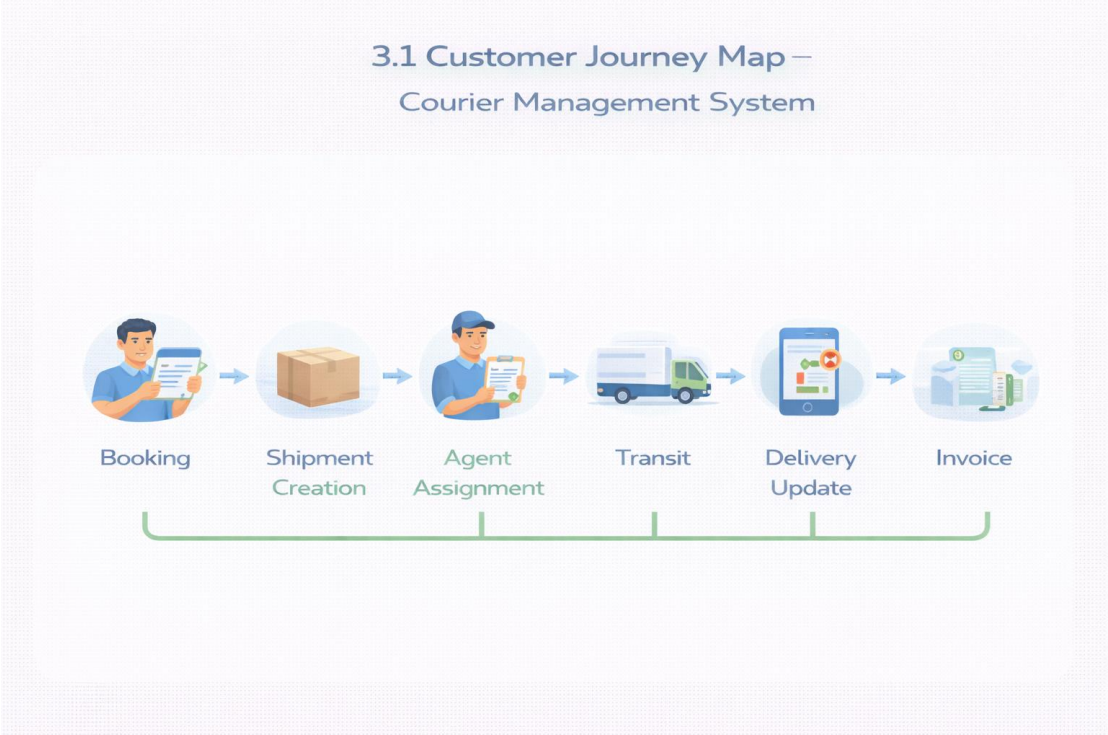
3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

The courier process follows these major stages:

Customer Booking → Shipment Creation → Data Validation → Agent Assignment → Shipment Transit → Delivery Status Update → Invoice Processing → Reporting

At each stage, Salesforce stores and manages the required information so that authorized users can monitor the shipment lifecycle.



3.2 Solution Requirements

Functional Requirements

FR No.	Requirement
FR-1	Customer management
FR-2	Shipment booking
FR-3	Delivery agent assignment
FR-4	Shipment status updates
FR-5	Invoice generation
FR-6	Reports and dashboards
FR-7	Branch management
FR-8	Delivery update management

Non-Functional Requirements

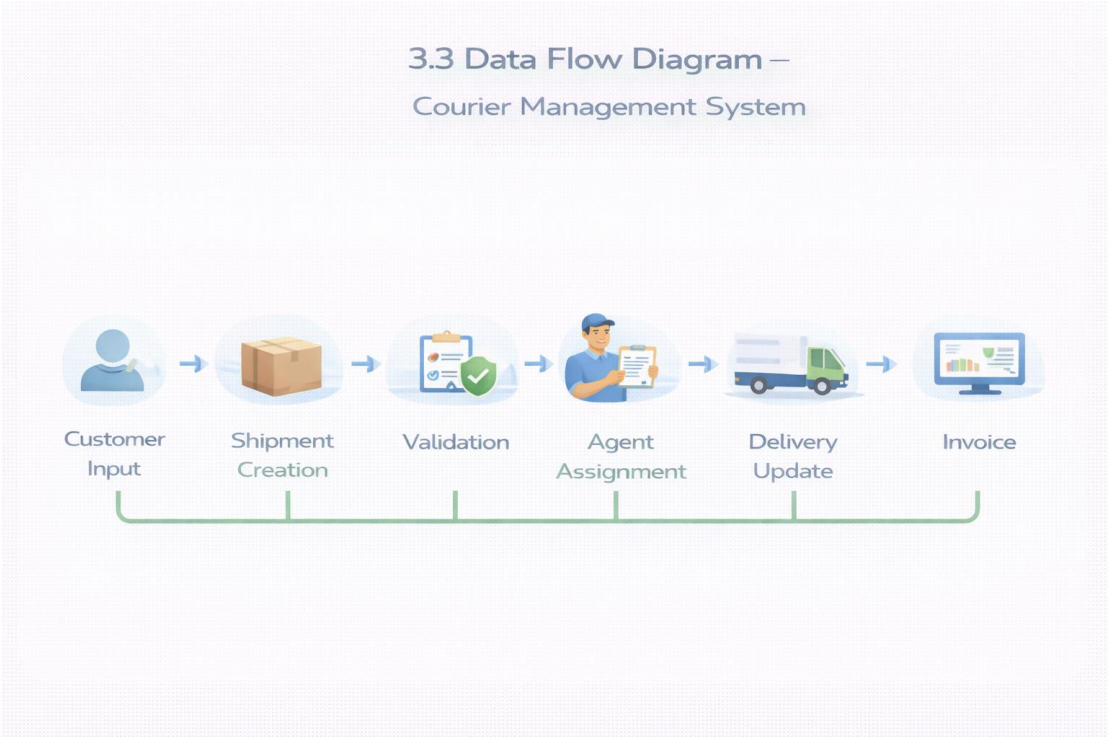
Requirement	Description
Usability	Simple Salesforce Lightning interface
Security	Role-based access and permissions
Reliability	Consistent and dependable automation
Performance	Quick shipment record and status updates
Availability	Cloud-based access
Scalability	Support for additional branches and users

3.3 Data Flow Diagram

The basic system flow is:

Customer Information → Shipment Creation → Data Validation → Delivery Agent Assignment → Shipment Tracking → Delivery Update → Invoice → Reports & Dashboards

Salesforce acts as the central platform for storing and processing the information throughout the workflow.



3.4 Technology Stack

Layer	Technology
User Interface	Salesforce Lightning
Application Logic	Salesforce Flows
Database	Salesforce Objects
Data Validation	Validation Rules
Security	Profiles and Roles
Reporting	Salesforce Reports
Visualization	Salesforce Dashboards
Cloud Platform	Salesforce

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The CMS addresses fragmented courier operations by bringing shipment, customer, delivery agent, branch, invoice, and delivery information into a centralized Salesforce environment.

Manual processes are reduced through Salesforce automation, while validation rules improve data quality. Shipment tracking provides better visibility, and reports and dashboards help managers understand operational performance.



4.2 Proposed Solution

Parameter	Description
Existing Problem	Manual and disconnected courier management
Proposed Solution	Salesforce-based Courier Management System
Main Innovation	Automated shipment tracking and workflow management
Business Impact	Faster and more organized courier operations
Business Model	Courier and logistics services
Scalability	Supports multiple branches

4.3 Solution Architecture

The proposed Salesforce solution contains the following components:

Customer Object – Maintains customer information.

Shipment Object – Stores parcel and tracking information.

Delivery Agent Object – Maintains delivery agent details.

Branch Object – Stores branch information.

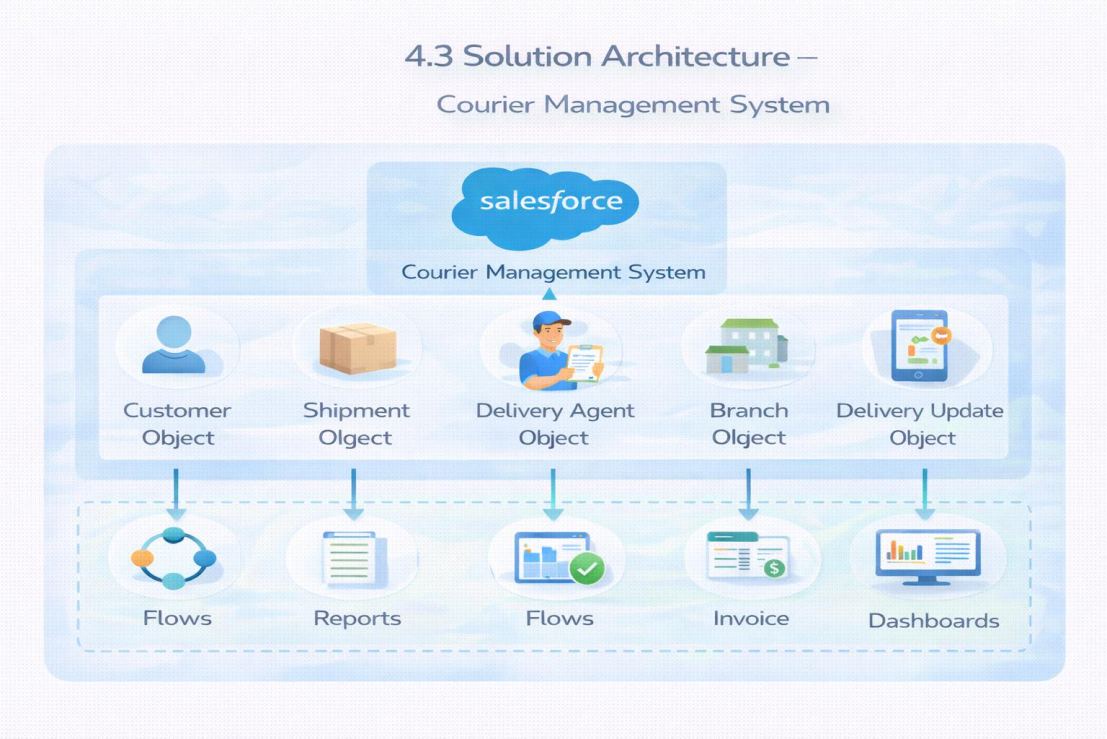
Delivery Update Object – Records shipment status changes.

Invoice Object – Maintains billing information.

Salesforce Flows – Automates business processes.

Reports – Provides detailed operational analysis.

Dashboards – Presents important information visually.



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project can be developed using an Agile methodology. Development activities are divided into smaller modules and implemented through multiple sprints.

The planning process includes:

Requirement identification

User story creation

Story point estimation

Sprint planning

Salesforce configuration

Testing

Review and deployment

Product Backlog and Sprint Schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Courier Management System.	3	High	Sachin.P
Sprint-2	Data Modeling	USN-2	As an admin, I want to create Customer, Shipment, Delivery Agent, Branch, and Invoice objects with proper relationships so that courier data is stored and managed efficiently.	5	High	Saran.S
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all courier modules so that I can easily navigate and perform daily operations.	5	High	Rogan Kennady.A
Sprint-3	Automation	USN-4	As a system, I want to validate shipment details such as delivery dates, addresses, and mandatory fields so that incorrect or incomplete	3	High	Ruppesh.E

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
			records are prevented.			
Sprint-3	Automation	USN-5	As a dispatcher, I want automated flows for shipment status updates and agent assignment so that deliveries are processed faster without manual intervention.	3	High	Prathap.E
Sprint-4	Security	USN-6	As an administrator, I want role-based access control for managers, agents, and staff so that sensitive shipment and billing data is secure.	5	High	Sachin.P
Sprint-5	Reports	USN-7	As a branch manager, I want reports showing delivery performance, shipment status, and revenue so that I can monitor operations and make informed decisions.	4	High	Ruppesh.E
Sprint-6	Dashboards	USN-8	As a branch manager, I want dashboards showing delivery performance, shipment status, and revenue so that I can monitor operations and make informed decisions.	4	Medium	Rogan kennady.A

Project Tracker

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	02 SEP 2026	07 SEP 2026	20	07 SEP 2026
Sprint-2	20	6 Days	09 SEP 2026	14 SEP 2026	20	14 SEP 2026
Sprint-3	20	6 Days	15 SEP 2026	20 SEP 2026	20	20 SEP 2026
Sprint-4	20	6 Days	22 SEP 2026	29 SEP 2026	20	29 SEP 2026

6. PROJECT DEVELOPMENT PHASE

What You'll Learn

Salesforce project implementation

Custom object creation

Object relationships

Page layout configuration

Validation rule development

Salesforce Flow automation

Reports and dashboard creation

Profiles and role configuration

User management

Application testing

Milestone 1: Salesforce Account

Introduction

Salesforce provides a cloud-based platform for developing business applications without requiring a traditional standalone database and server infrastructure.

For the CMS project, Salesforce is used to organize courier information, automate operational activities, control user access, and provide business reporting.

What is Salesforce?

Salesforce is a cloud-based customer relationship and business application platform that provides tools for managing data, automating workflows, creating reports, and controlling user access. In the CMS project, these capabilities are applied to courier and logistics operations.

Activity 1: Creating Developer Account

A Salesforce Developer Account is created to develop and test the CMS.

General steps:

Open the Salesforce Developer signup page.

Enter first name and last name.

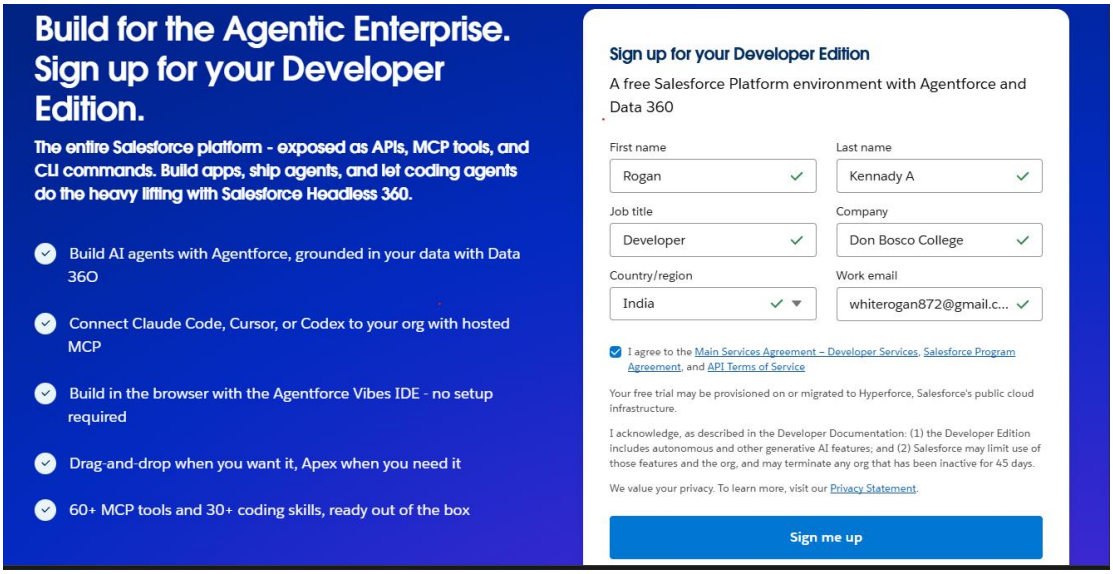
Enter email address.

Select Developer as the job title.

Enter the college or organization name.

Select India as the country.

Submit the registration form.



Activity 2: Account Activation

Open the registered email account.

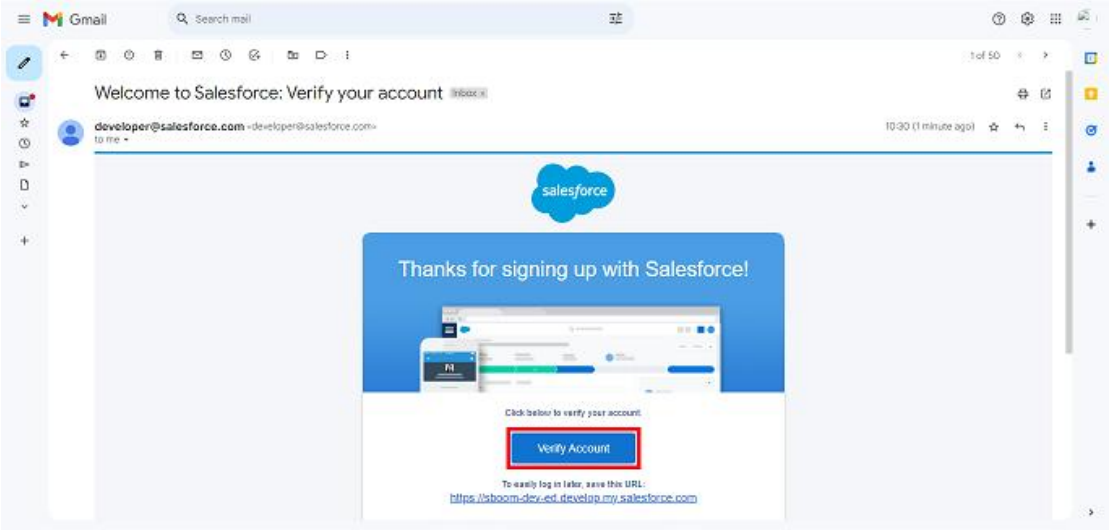
Open the Salesforce verification email.

Select the verification option.

Create the Salesforce password.

Complete the security information.

Access the Salesforce setup environment.



Milestone 2 – Objects

What is an Object?

Salesforce Objects are used to store business information. An object contains fields that define the type of information and records that contain the actual data.

Custom Objects Created

Customer__c – Stores customer details.

Shipment__c – Stores parcel and shipment information.

Delivery_Agent__c – Maintains delivery agent details.

Branch__c – Represents courier branches.

Delivery_Update__c – Stores shipment status updates.

Invoice__c – Maintains billing and payment information.

Relationships

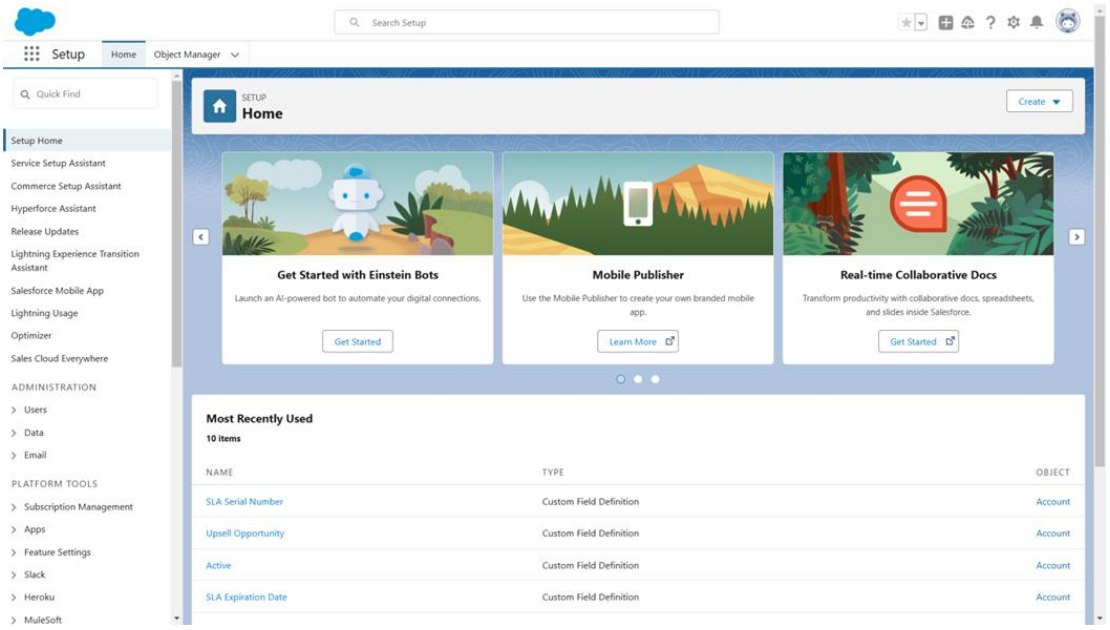
Customer → Shipment — Lookup

Shipment → Delivery Agent — Lookup

Delivery Agent → Branch — Lookup

Delivery Update → Shipment — Master-Detail

Invoice → Shipment — Lookup



Activity 1: Create Customer Object

Open Setup.

Select Object Manager.

Select Create → Custom Object.

Enter Label: Customer.

Enter Plural Label: Customers.

Set Record Name as Customer Name.

Enable reporting and field history if required.

Save the object.

Activity 2: Create Shipment Object

Open Setup → Object Manager.

Select Create → Custom Object.

Label: Shipment.

Plural Label: Shipments.

Record Name: Tracking Number.

Select Auto Number.

Example format: SHP-{00000}.

Enable reporting and field history.

Save.

Activity 3: Create Delivery Agent Object

Create the Delivery Agent custom object with Agent Name as the record name and enable reporting and field history.

Activity 4: Create Branch Object

Create the Branch custom object with Branch Name as the record name and enable reporting.

Activity 5: Create Delivery Update Object

Create the Delivery Update object using an Auto Number record name such as:

UPD-{0000}

Activity 6: Create Invoice Object

Create the Invoice object using an Auto Number record name such as:

INV-{0000}

Milestone 3 – Tabs

Salesforce Tabs provide users with direct access to object records from the Lightning application.

Create tabs for:

Customer

Shipment

Delivery Agent

Branch

Delivery Update

Invoice

For each object, open Setup → Tabs → New, select the required object, choose a tab style, and save the configuration.

Milestone4 - Fields & Relationships

What are Fields?

Fields represent individual data items stored inside Salesforce records. They are similar to columns in a relational database.

Customer Fields

Field	Data Type	Required
Customer Name	Text	Yes
Email	Email	Yes
Phone	Phone	Yes
Address	Text Area	Yes
Customer Type	Picklist	Yes
City	Text	No
State	Text	No
Pincode	Text	No
Is Active	Checkbox	No

Shipment Fields

Field	Data Type	Required
Tracking Number	Auto Number	Yes
Customer	Lookup	Yes
Source Address	Text Area	Yes
Destination Address	Text Area	Yes
Weight	Number	Yes
Price	Currency	Yes
Status	Picklist	Yes
Expected Date	Date	Yes
Assigned Agent	Lookup	No
Shipment Type	Picklist	Yes

Shipment Status Values

Booked

In Transit

Out for Delivery

Delivered

Cancelled

Shipment Type Values

Domestic

International

Delivery Agent Fields

Agent Name

Email

Phone

Branch

Status

Joining Date

Delivery Agent Status

Active

Inactive

On Leave

Branch Fields

Branch Name

Branch Code

City

State

Manager Name

Delivery Update Fields

Update Number

Shipment

Status

Update Date

Remarks

Delivery Update Status

In Transit

Out for Delivery

Delivered

Delayed

Invoice Fields

Invoice Number

Shipment

Amount

Invoice Date

Payment Status

Payment Status

Paid

Unpaid

Pending

Relationships

Child Object	Field	Parent Object	Relationship
Shipment	Customer	Customer	Lookup
Shipment	Assigned Agent	Delivery Agent	Lookup
Delivery Agent	Branch	Branch	Lookup
Delivery Update	Shipment	Shipment	Master-Detail
Invoice	Shipment	Shipment	Lookup

Milestone 5 – Validation Rules

Validation Rules ensure that users cannot save records containing invalid or incomplete information.

Customer Validation

Email Required

ISBLANK(Email__c)

Phone Validation

LEN(Phone__c) <> 10

Pincode Validation

LEN(Pincode__c) <> 6

Shipment Validation

Weight Validation

Weight__c <= 0

Price Validation

Price__c <= 0

Expected Date Validation

Expected_Date__c < TODAY()

Delivery Agent Validation

Email Validation

NOT(CONTAINS(Email__c, "@"))

Branch Validation

Branch Code Required

ISBLANK(Branch_Code__c)

Delivery Update Validation

If a shipment is marked as delayed, remarks must be entered:

```
AND(
   ISPICKVAL(Status__c, "Delayed"),
   ISBLANK(Remarks__c)
)
```

Milestone 6 – Page Layouts

Page Layouts determine how fields, sections, buttons, and related information are displayed on Salesforce record pages.

Customer Layout

Customer Information

- Customer Name
- Customer Type
- Email
- Phone
- Is Active

Address Information

- Address
- City
- State
- Pincode

Shipment Layout

Shipment Details

- Tracking Number
- Customer
- Shipment Type
- Weight
- Price
- Status

Address Details

- Source Address
- Destination Address

Delivery Details

- Expected Date
- Assigned Agent

Delivery Agent Layout

- Agent Name
- Email
- Phone
- Branch
- Status
- Joining Date

Branch Layout

- Branch Name
- Branch Code
- City
- State
- Manager Name

Delivery Update Layout

- Update Number
- Shipment
- Status
- Update Date
- Remarks

Invoice Layout

Invoice Number

Shipment

Amount

Invoice Date

Payment Status

Milestone 7 – Flows

Salesforce Flow is used to automate business processes without requiring traditional programming.

Activity 1: Automatic Shipment Status Update

Scenario

When a new Delivery_Update__c record is created, the related shipment should automatically receive the same status.

Flow Configuration

Object: Delivery_Update__c

Trigger: Record is created

Action: Update related Shipment__c

The flow identifies the shipment connected to the delivery update and copies the latest status into the Shipment record.

Flow Process

Delivery Update Created → Find Related Shipment → Update Shipment Status → Save

After creating the flow:

Save the flow.

Give it a meaningful name.

Check the configuration.

Activate the flow.

Milestone 8 – The Lightning App

What is an App?

A Salesforce App combines related tabs and features into a single workspace so that users can access the required modules easily.

Activity 1: Create Lightning App

Create:

Courier Management System App

Include:

- Customers
- Shipments
- Delivery Agents
- Branches
- Delivery Updates
- Invoices
- Reports
- Dashboards

The app can then be assigned to authorized Salesforce profiles.

Milestone 9 – Profiles

What is a Profile?

A Salesforce Profile controls the permissions and access available to a user. It can determine access to objects, fields, applications, tabs, and other Salesforce features.

Profiles Created

Profile	Purpose
Courier Admin	Complete CMS administration

Profile	Purpose
Branch Manager	Manage branch operations
Delivery Agent	Update assigned deliveries
Customer Support	Handle customer and shipment information

Courier Admin

The Courier Admin profile provides administrative access for configuring and managing the CMS.

Branch Manager

The Branch Manager can work with:

- Customers
- Shipments
- Delivery Agents
- Branches
- Delivery Updates
- Invoices
- Reports
- Dashboards

Delivery Agent

The Delivery Agent mainly works with:

- Shipments
- Delivery Updates

The agent has limited access to administrative functions.

Customer Support

Customer Support can access:

- Customers
- Shipments

Delivery Updates

Invoice information

Milestone 10 – Role

Salesforce Roles establish the organizational hierarchy and influence record visibility.

Role Hierarchy

Courier Admin
↓
Branch Manager
↓
Delivery Agent

Branch Manager
↓
Customer Support

Purpose

Courier Admin: Controls overall system operations.

Branch Manager: Supervises branch-level activities.

Delivery Agent: Handles assigned shipment deliveries.

Customer Support: Assists customers with shipment-related information.

Milestone 11 – User

A Salesforce User is an individual who accesses the Salesforce organization using a valid user account.

Each user should have appropriate:

Username

Email

Role

Profile

License

User Types

User Type	Role	Profile
Courier Admin	Courier Admin	Courier Admin
Branch Manager	Branch Manager	Branch Manager
Delivery Agent	Delivery Agent	Delivery Agent
Customer Support	Customer Support	Customer Support

Users should be assigned permissions according to their responsibilities.

Milestone15 –Reports & Dashboards

Reports and Dashboards provide management with a visual and analytical view of courier operations.

Activity 1: Create Report Types

Shipment Report Type

Primary Object: Shipment__c

Label: Shipment Tracking Report

Customer Report Type

Primary Object: Customer__c

Label: Customer Summary Report

Agent Report Type

Primary Object: Delivery_Agent__c

Label: Agent Performance Report

Activity 2: Create Reports

Report 1 – Shipment Status Summary

Include:

Tracking Number

Customer

Status

Expected Date

Assigned Agent

Group the report using shipment status and summarize shipment charges where required.

Report 2 – Agent Performance Report

Include:

Agent Name

Branch

Shipment Count

Shipment Status

Group the information according to delivery agent.

Report 3 – Revenue by Branch

Include:

Tracking Number

Customer

Price

Invoice Status

Group the results by:

Branch

Month

Activity 3: Create Dashboard

Create:

Courier Management Dashboard

Dashboard Components

Component 1 – Shipment Status Overview

Use a Donut Chart to display shipment status.

Component 2 – Agent Performance

Use a Bar Chart to compare delivery activities.

Component 3 – Revenue by Branch

Use a Column Chart to display branch-level revenue information.

The dashboard can be shared with authorized users such as Courier Admin, Branch Managers, and Customer Support.

7.FUNCTIONAL & PERFORMANCE TESTING

The developed Salesforce CMS should be tested to verify that all major functions operate correctly.

Testing includes:

- Customer record creation
- Shipment record creation
- Required-field validation
- Phone and pincode validation
- Shipment status updates
- Delivery agent assignment
- Flow automation
- Invoice creation
- Profile-based access
- Role-based visibility
- Report generation
- Dashboard visualization

The testing process ensures that the configured Salesforce components work together correctly and that users can perform their assigned activities.

8. ADVANTAGES & DISADVANTAGES

Advantages

Centralized courier information

Reduced manual processing

Automated business workflows

Better shipment visibility

Improved customer communication

Multiple branch support

Role-based security

Real-time reports and dashboards

Improved operational monitoring

Cloud-based accessibility

Disadvantages

Dependence on the Salesforce platform

Initial configuration requires effort

Some advanced Salesforce features may require additional licensing

Users require training to operate the system effectively

9. CONCLUSION

The Courier Management System provides a centralized Salesforce-based approach for organizing courier and logistics activities. The system brings customer management, shipment processing, delivery agent management, branch operations, delivery updates, and billing into a unified cloud environment.

Salesforce automation reduces repetitive manual work, while validation rules improve data accuracy. Reports and dashboards help managers monitor shipment activity and operational performance. The system therefore provides a structured approach to courier management while improving visibility, coordination, and service efficiency.

10. FUTURE SCOPE

The CMS can be enhanced in the future with additional technologies and features such as:

Mobile-based shipment tracking

GPS-based delivery monitoring

SMS and WhatsApp notifications

AI-powered delivery route optimization

Predictive delivery-time estimation

Customer self-service tracking portal

Digital payment integration

Automated customer feedback

Advanced business intelligence dashboards

Integration with external logistics systems

11. APPENDIX

Source Code

Salesforce configuration, custom objects, fields, relationships, validation rules, flows, profiles, roles, reports, and dashboards.

Dataset

Courier information maintained through Salesforce Objects, including:

Customer records

Shipment records

Delivery Agent records

Branch records

Delivery Update records

Invoice records

Demo

Salesforce Developer Organization / Salesforce CMS Application

Project Technology

Platform: Salesforce
Interface: Salesforce Lightning
Automation: Salesforce Flow
Database: Salesforce Objects
Security: Profiles and Roles
Reporting: Salesforce Reports and Dashboards

Final Statement

The Courier Management System demonstrates how Salesforce can be used to transform traditional courier operations into a centralized, automated, and cloud-based business solution.

This version keeps the same major titles and topics from your uploaded text, but the actual explanations and paragraphs are rewritten with different wording.